

Decent Coaching Classes

VIII SB

Mind Your Math - VII

Answer Key & Marking Scheme

Marks: 40

Time: 2 Hrs.

Q.1 (A) Correct alternatives

[5]

- (1) (C) $\sqrt{7}$ [1]
- (2) (B) 115° (interior angles on the same side of the transversal are supplementary: $180^\circ - 65^\circ = 115^\circ$) [1]
- (3) (D) 0.2 ($0.2 \times 0.2 \times 0.2 = 0.008$) [1]
- (4) (B) the vertex of the right angle [1]
- (5) (C) ₹450 (discount = 10% of 500 = ₹50; $500 - 50 = ₹450$) [1]

Q.1 (B) Solutions

[3]

- (1) $7/8 = 0.875$ [1]
- (2) $A \propto r^2$ [1]
- (3) Trapezium [1]

Q.2 Any four (2 marks each)

[8]

- (a) First compare $3/4$ and $5/6$: $3/4 = 9/12$, $5/6 = 10/12$. $9/12 < 10/12$, so $3/4 < 5/6$. [1]
For negative numbers the order reverses: $-3/4 > -5/6$. [1]
- (b) $(m + 7)(m - 3) = m^2 + (7 - 3)m + (7)(-3)$ [1]
 $= m^2 + 4m - 21$ [1]
- (c) Factors of 20 whose sum is 9 are 4 and 5. [1]
 $x^2 + 9x + 20 = x^2 + 4x + 5x + 20 = x(x + 4) + 5(x + 4) = (x + 4)(x + 5)$ [1]
- (d) $3375 = 3 \times 3 \times 3 \times 5 \times 5 \times 5 = (3 \times 5)^3 = 15^3$ [1]
Cube root of 3375 = 15 [1]
- (e) In the right angled triangle formed by the diagonal: $(\text{other side})^2 = 13^2 - 5^2 = 169 - 25 = 144$ [1]
Other side = $\sqrt{144} = 12$ cm [1]
- (f) Commission = $4,50,000 \times 2/100 = ₹9,000$ [1]
Amount received by the seller = $4,50,000 - 9,000 = ₹4,41,000$ [1]

Q.3 Any four (3 marks each)

[12]

- (a) Interior angles on the same side of the transversal are supplementary.
 $(2x + 10) + (3x + 20) = 180$ [1]
 $5x + 30 = 180$, $5x = 150$, $x = 30$ [1]
Angles: $2(30) + 10 = 70^\circ$ and $3(30) + 20 = 110^\circ$ [1]
- (b) $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$ [1]
 $16^{3/4} = (16^{1/4})^3 = 2^3 = 8$ [1]
Value = $4 + 8 = 12$ [1]
- (c) The centroid divides each median in the ratio 2 : 1.
 $l(AG) = 2 \times l(GD) \rightarrow l(GD) = 10/2 = 5$ cm [1]
 $l(BG) = 2 \times l(GE) = 2 \times 4 = 8$ cm [1]
 $l(BE) = l(BG) + l(GE) = 8 + 4 = 12$ cm [1]
- (d) Using $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ with $a = 3x$, $b = 2y$ [1]

$$= (3x)^3 + 3(3x)^2(2y) + 3(3x)(2y)^2 + (2y)^3 \quad [1]$$

$$= 27x^3 + 54x^2y + 36xy^2 + 8y^3 \quad [1]$$

- (e) Product of first and last coefficients = $2 \times (-12) = -24$; factors with sum -5 are -8 and 3 . $[1]$

$$2x^2 - 5x - 12 = 2x^2 - 8x + 3x - 12 \quad [1]$$

$$= 2x(x - 4) + 3(x - 4) = (x - 4)(2x + 3) \quad [1]$$

- (f) $x \propto y$, so $x = ky$. $8 = k \times 20 \rightarrow k = 2/5$ (constant of variation) $[1]$

$$\text{Equation of variation: } x = (2/5)y \quad [1]$$

$$\text{When } y = 35: x = (2/5) \times 35 = 14 \quad [1]$$

Q.4 Any three (4 marks each)

[12]

- (a) Steps of construction:

1. Draw a number line; mark O (0) and A (2). $[1]$

2. At A draw a line l perpendicular to the number line; on it mark point P such that $AP = 1$ unit. $[1]$

3. Join OP. In right angled $\triangle OAP$, $OP^2 = OA^2 + AP^2 = 2^2 + 1^2 = 5$, so $OP = \sqrt{5}$. $[1]$

4. With centre O and radius OP draw an arc cutting the number line to the right of O at Q. Point Q shows $\sqrt{5}$. $[1]$

- (b) Rough figure with given measures shown $[1]$

Seg $MO = 5$ cm drawn; ray at M making 75° and ray at O making 105° drawn $[1]$

Point R marked on the ray from O with $l(OR) = 4$ cm; ray from R making 90° with RO drawn on the side of M $[1]$

The two rays meet at E, giving $\square MORE$ ($m\angle E = 360^\circ - (75^\circ + 105^\circ + 90^\circ) = 90^\circ$). Approx. $l(ME) \approx 5.3$ cm, $l(RE) \approx 4.8$ cm (accept ± 0.2 cm). $[1]$

- (c) (i) $8x^3 + 125y^3 = (2x)^3 + (5y)^3 \quad [1/2]$

$$= (2x + 5y)[(2x)^2 - (2x)(5y) + (5y)^2] = (2x + 5y)(4x^2 - 10xy + 25y^2) \quad [1\frac{1}{2}]$$

(ii) $6 \times (-15) = -90$; factors with sum -1 are -10 and 9 . $[1]$

$$6x^2 - x - 15 = 6x^2 - 10x + 9x - 15 = 2x(3x - 5) + 3(3x - 5) = (3x - 5)(2x + 3) \quad [1]$$

- (d) Discount = 12% of $2500 = 2500 \times 12/100 = ₹300 \quad [1]$

$$\text{Selling price} = 2500 - 300 = ₹2,200 \quad [1]$$

Profit 10% : if cost price is ₹100, selling price is ₹110, so $CP/SP = 100/110 \quad [1]$

$$\text{Cost price} = 2200 \times 100/110 = ₹2,000 \quad [1]$$

- (e) More workers means fewer days, so it is inverse variation: $w \propto 1/d$, i.e. $w \times d = k$. $[1]$

$$k = 12 \times 15 = 180 \quad [1]$$

$$w \times 9 = 180 \quad [1]$$

$$w = 20. \text{ 20 workers are required. } [1]$$